Project Title - SentimentEdge: A Multi-Source Stock Sentiment Intelligence Platform

Agentic Systems Studio - Phase 1: Scoping and Justification Track A - Technical Build

Team Members

Farrukh, Pablo, Moid, Afaq

Selected Track

Track A - Technical Build

1. One-Paragraph Project Summary

SentimentEdge is an agentic stock market sentiment intelligence platform. It is designed to monitor social media platforms for stock sentiment and insider chatter. The goal is to give retail investors access to market signals that have traditionally only been available to well-connected institutional players. The system is architected to work across multiple social media platforms including Reddit, X, StockTwits, and others. For the scope of this project it will be implemented on Reddit finance communities. Using a five-agent architecture, SentimentEdge transforms raw social media text into ticker-level sentiment scores, emotion breakdowns, and sarcasm-corrected signals. It also detects unconfirmed merger and acquisition rumours, corporate deal chatter, and insider speculation that surfaces on social platforms before reaching mainstream financial news. The primary use case is a retail investor who types in a ticker symbol and receives a sentiment report and any active rumour alerts in under 30 seconds, sourced entirely from public social media.

---

2. Target User

Primary User: Retail investors and finance students who follow Reddit communities like r/wallstreetbets, r/investing, and r/stocks. They lack the time or tools to manually aggregate and interpret sentiment signals across thousands of posts.

User Pain Points:

- Sentiment is scattered across thousands of posts and is impossible to read manually
- Sarcasm, memes, and irony make social media text misleading at face value
- There is no easy way to track how sentiment for a specific ticker changes over time
- Existing tools either aggregate pre-scored sentiment with no transparency or require expensive API access
- By the time a merger or acquisition is confirmed publicly, institutional investors have already acted on early signals. Retail investors are always last to know.
- Insider chatter and unconfirmed rumours about major deals often circulate on social media days or weeks before official announcements. Retail investors have no systematic way to detect these signals.

User Goal: Get a clear, honest, confidence-scored sentiment signal for any stock ticker sourced directly from social media. Also receive early warnings about unconfirmed deal rumours and insider chatter before these stories reach mainstream financial news. All of this in under 30 seconds without reading hundreds of posts manually.

Primary Use Case: A retail investor types in a ticker symbol and receives a sentiment report and any active rumour alerts in under 30 seconds, sourced entirely from public social media.

---

3. Problem Statement

Reddit finance communities generate thousands of posts daily about individual stocks. These posts contain genuine market signal including real investor positions, reactions to earnings, and sentiment shifts. However, this signal is buried under noise including deleted posts, memes, sarcasm, and off-topic discussions. A retail investor who wants to know what the community thinks about $TSLA today would need to read hundreds of posts. They would need to mentally filter out the noise, detect sarcasm, and aggregate the remaining signals into a coherent view. This is not feasible.

Current solutions either provide pre-aggregated scores with no transparency (StockTwits, ApeWisdom), require enterprise-level API costs (X/Twitter at $200 or more per month), or are too general to handle the unique language of finance communities. SentimentEdge addresses this gap by processing raw text through a transparent, auditable multi-agent pipeline. It explicitly handles sarcasm, filters noise by post quality, and produces confidence-scored insights with direct evidence links.

A second and more significant problem exists beyond general sentiment. Social media platforms regularly surface unconfirmed insider chatter about major corporate events including mergers, acquisitions, partnerships, and leadership changes. This chatter often appears days or weeks before official announcements. Well-connected institutional investors already have access to this information through professional networks. Retail investors do not. This information asymmetry is a fundamental unfairness in financial markets. SentimentEdge addresses this directly by detecting and flagging posts that contain deal rumours or unusual corporate chatter. It surfaces these signals for retail investors at the same time they appear publicly on social media, not weeks later when the price has already moved.

---

4. Why This is an Agentic Problem

A single prompt to a single AI model cannot solve this problem reliably. The task requires multiple distinct processing steps. Each step has different inputs, outputs, decision logic, and failure modes. This is a multi-step reasoning and coordination problem, not a simple retrieval or summarization task.

A single-agent approach would either skip the filtering step and waste API budget analyzing deleted posts, or skip the LLM step entirely and miss sarcasm. Only a coordinated pipeline handles all failure modes reliably.

Why each step needs its own agent:

- **Data collection** - Requires API calls, pagination, and rate limit handling. This is an engineering task, not a reasoning task. A single LLM prompt cannot pull live data from Reddit.

- **Noise filtering** - Requires rule-based logic on metadata including score, flair, and deleted status. In our dataset, 64% of raw posts were deleted or removed. This must be handled before any AI analysis.
- **Sentiment analysis** - Requires LLM reasoning to handle sarcasm, finance slang, meme language, and ticker extraction. It also requires a separate detection layer for insider chatter and deal rumours. This is where Claude is genuinely needed.
- **Insider chatter detection** - Identifying posts that contain unconfirmed deal rumours requires a different kind of analysis than general sentiment scoring. A dedicated detection layer is needed to flag these posts, assess their credibility, and route them to a separate output track.
- **Signal aggregation** - Requires weighted averaging across hundreds of posts grouped by ticker and time period. This is a data processing task, not an AI task.
- **Output formatting** - Requires structuring two separate output tracks: a general sentiment report and a rumour alert track. It also requires appropriate warnings, confidence indicators, and source links.

Evidence from prototyping confirms this requirement. During scoping, 11.8% of analyzed posts were identified as sarcastic by Claude. A rule-based sentiment tool would have badly misclassified these. The highest-scoring post in the dataset had a score of 123,150 but a confidence score of only 0.30 because it was a viral meme with minimal market content. These failure cases cannot be handled by a single prompt. They require a dedicated filtering stage before text reaches the sentiment layer and a dedicated confidence-scoring step after it.

---

5. Initial Architecture Concept

The system is designed as a platform-agnostic pipeline. The Collector Agent is built to interface with any social media source. For this project it is implemented on Reddit. Future versions would extend the same pipeline to X, StockTwits, LinkedIn, and other platforms without changing the downstream agents.

The system follows a sequential pipeline architecture with five agents:

Collector Agent -> Filter Agent -> Sentiment Agent -> Aggregator Agent -> Output Agent

Data flow:

- Social media platforms (Reddit in scope, X and StockTwits as future extensions) feed into the Collector Agent
- Collector Agent passes a raw post dataframe to the Filter Agent
- Filter Agent passes clean post text to the Sentiment Agent
- Sentiment Agent passes per-post JSON scores to the Aggregator Agent
- Aggregator Agent passes ticker-level summaries and rumour alerts to the Output Agent
- Output Agent delivers two tracks to the user: a general sentiment report and a rumour alert track

Coordination logic:

- The pipeline runs sequentially. Each agent waits for the previous to complete.
- The Filter Agent acts as a gate. Low quality posts never reach the LLM.
- The Sentiment Agent processes the top 500 posts by engagement score to control API costs.
- The Aggregator flags tickers with fewer than 5 posts as low-confidence.
- The Aggregator routes posts with rumour_confidence above 0.7 to the separate rumour alert track.
- The Output Agent displays warnings when confidence is below 0.6 or sarcasm rate exceeds 10%.
- The pipeline stops when the Output Agent delivers the report or when an agent returns zero results after filtering.

---

6. Multi-Agent System Canvas (Infographic attached at the end)

System Name: SentimentEdge

What the system does: Transforms raw social media text into structured, ticker-level stock sentiment insights and insider chatter alerts through a coordinated five-agent pipeline.

Inputs: Social media platform configuration, subreddit names, date range, ticker query from user

Outputs: Two output tracks. Track 1 is a ticker sentiment report including overall sentiment score, confidence, emotion breakdown, and monthly timeline. Track 2 is a rumour and insider chatter alert flagging unconfirmed deal speculation detected in the data.

Agents involved: Collector, Filter, Sentiment Analyzer, Aggregator, Output

Coordination pattern: Sequential pipeline. Each agent passes structured output to the next.

Human-in-the-loop points: The user queries a specific ticker at the Output stage. Low-confidence outputs surface a warning prompting the user to verify manually. Rumour alerts always display the source post so users can judge credibility themselves.

Stopping condition: The pipeline completes when the Output Agent delivers the formatted report. It also stops when an agent returns zero results after filtering.

---

7. Agent Canvases

Agent 1 - Collector Agent

Role: Retrieve raw posts from a configured social media source and standardize into a consistent schema. For this project the source is Reddit.

Inputs: Platform configuration, subreddit names, date range, post limit, optional ticker keyword filter

Outputs: Raw post dataframe with columns: id, title, body, score, upvote_ratio, num_comments, flair, author, created_utc, permalink, subreddit

Tools / APIs: Reddit API via PRAW for live data, Arctic Shift for historical data, pandas

Decision logic: Paginate through subreddit feeds. Collect from r/wallstreetbets, r/investing, r/stocks, r/options, and r/stockmarket. Stop when the post limit is reached or the rate limit is hit. PRAW handles rate limiting automatically at 100 requests per minute.

Agent 2 - Filter Agent

Role: Remove low-quality, irrelevant, and deleted posts. Retain only posts with meaningful market signal and enrich with derived columns.

Inputs: Raw post dataframe from Collector Agent

Outputs: Filtered dataframe with added columns: is_deleted flag, engagement_score, is_relevant flag, combined text column

Tools / APIs: pandas, regex for ticker extraction

Decision logic:

- Remove posts where selftext is [removed] or [deleted]
- Remove posts with score below 10
- Remove posts with no ticker mention or finance keyword
- Compute engagement_score as (score x 0.7) plus (num_comments x 0.3)
- Prioritise gain, loss, YOLO, news, and DD flairs over meme and other

---

Agent 3 - Sentiment Agent

Role: Analyze each post using Claude to extract sentiment, emotions, sarcasm, ticker, and insider chatter signals.

Inputs: Clean text per post (title plus body, max 500 characters), score, num_comments, flair

Outputs: Per-post JSON including sentiment, confidence, tickers, primary_ticker, emotions, post_type, is_sarcastic, market_relevance, key_insight, is_rumour, rumour_type, rumour_confidence, and rumour_summary

Tools / APIs: Anthropic Claude API (claude-sonnet-4-20250514), JSON parsing

Decision logic: Send a structured prompt to Claude for each post. Parse the JSON response. On parse failure return neutral with confidence 0.0. Apply a 0.3 second sleep between calls for rate limiting. Process the top 500 posts by engagement score per run to control API costs at approximately $0.50 per run. In addition to sentiment, Claude is prompted to detect unconfirmed deal chatter. Rumour types include merger speculation, acquisition rumour, partnership chatter, leadership change, and regulatory decision.

Agent 4 - Aggregator Agent

Role: Combine per-post sentiment scores into ticker-level summaries with engagement weighting. Route high-confidence rumour posts to a separate alert track.

Inputs: Per-post sentiment dataframe from Sentiment Agent

Outputs: Two outputs. First, a ticker summary dataframe with ticker, post_count, bullish_pct, bearish_pct, neutral_pct, avg_confidence, dominant_emotions, sarcastic_count, and monthly_trend. Second, a rumour alert list for posts with rumour_confidence above 0.7.

Tools / APIs: pandas groupby, weighted averaging

Decision logic: Group by primary_ticker. Weight sentiment by engagement_score. Compute bullish, bearish, and neutral percentages. Flag tickers with fewer than 5 posts or average confidence below 0.6. Generate a monthly sentiment timeline per ticker. Route posts with rumour_confidence above 0.7 to the separate rumour alert track.

Agent 5 - Output Agent

Role: Format aggregated results into two structured outputs for the end user: a general sentiment report and a rumour alert track.

Inputs: Ticker summaries and rumour alerts from Aggregator Agent, user query (ticker symbol)

Outputs: Track 1 is a formatted sentiment report showing overall score, confidence, emotion breakdown, monthly timeline, post type breakdown, and top insights. Track 2 is a rumour alert showing the unconfirmed deal claim, the source post, the rumour type, and the rumour confidence score.

Tools / APIs: Python formatting, Streamlit dashboard planned for Phase 2

Decision logic: Retrieve ticker summary matching user query. Display a low-confidence warning if average confidence is below 0.6. Display a sarcasm-adjustment note if sarcastic posts exceed 10%. Always show the source post link so users can verify. For rumour alerts, always display a disclaimer that the source is unverified social media speculation and does not constitute financial or legal advice.

---

8. Initial Success Criteria

Criterion	Target	How We Measure It
Sentiment accuracy	Claude confidence >= 0.75 on high-quality posts	Average confidence score across analyzed posts
Sarcasm detection	Correctly flag >= 80% of sarcastic posts	Manual review of 50 flagged posts
Ticker extraction	>= 90% of posts correctly tagged to primary ticker	Manual spot-check of 100 posts
Pipeline throughput	Process 500 posts in under 10 minutes	End-to-end timing

User query response	Return ticker report in under 30 seconds	Response time measurement
Filter quality	Filtered dataset retains >= 30% of raw posts	Row count before and after filtering
Coverage	At least 10 tickers with >= 5 posts each	Ticker distribution table
Rumour detection	Flag at least 5 credible rumour posts per 500 analyzed	Count of posts with rumour_confidence above 0.7
Rumour detection precision	Less than 20% false positive rate on flagged rumour posts	Manual review of all flagged rumour alerts
Failure case documentation	At least 2 documented failure cases with explanation	Failure log with case, cause, and fix attempted

---

9. Scope Boundaries

In Scope:

- Reddit as the implemented data source for this project
- Platform-agnostic architecture designed for future multi-platform expansion to X, StockTwits, and others
- US equities only
- 5 finance subreddits: r/wallstreetbets, r/investing, r/stocks, r/options, r/stockmarket
- Sentiment scoring, emotion detection, sarcasm detection, post type classification, and ticker extraction
- Detection of unconfirmed merger rumours, acquisition chatter, and deal speculation
- Batch processing, not real-time streaming
- Historical data from April 2025 to March 2026
- Top 500 posts by engagement per run for LLM analysis
- Ticker-level and monthly sentiment aggregation

Out of Scope:

- X, StockTwits, and LinkedIn as live data sources (architecture supports them, implementation deferred to future phases)
- Verifying or confirming whether rumours are true. We surface signals, not facts.
- Legal or compliance screening of insider information
- Cryptocurrency assets

- International markets
- Real-time streaming pipeline
- Trading advice or price predictions
- Options pricing or technical analysis
- User authentication or personalisation
- Comment-level analysis in version 1

Note on rumour detection base rate: Rumour detection is expected to flag a small number of posts per run. Most runs will return zero rumour alerts. This is by design, not a failure. The system is built to surface genuine signals rarely rather than flood users with noise.

---

10. Initial Risks and Concerns

Sarcasm misclassification WSB uses heavy irony. In our prototype, 11.8% of posts were sarcastic. A simple sentiment tool would misread these badly. Mitigation: Use Claude with an explicit sarcasm flag. Display the confidence score to the user so they can make informed judgments.

Low-content viral posts The highest-scoring post in our dataset had 123,150 upvotes but a confidence score of only 0.30. It was a meme with no real market content. High engagement does not equal high signal. Mitigation: Filter by market_relevance score. Display a low-confidence warning when confidence falls below 0.6.

64% deleted posts The majority of raw posts have removed bodies. Only the title is available for these posts. This is a real data quality problem. Mitigation: Use title-only text as a fallback with a flag. Prioritise posts that have full body content.

API rate limits and cost Reddit API caps at 100 requests per minute. Claude API costs scale with volume. Mitigation: PRAW handles rate limits automatically. Process only the top 500 posts per run to control Claude API costs at approximately $0.50 per run.

Ticker misidentification Common words like OPEN, FAST, and AI are also valid stock tickers. This creates false positives. Mitigation: Prioritise the $TICKER format. Require financial context confirmation for ambiguous short tickers.

Rumour misidentification The system may flag speculative creative writing, hypothetical discussions, or jokes as genuine insider chatter. Mitigation: Require rumour_confidence above 0.7 before surfacing an alert. Show the source post so users can judge credibility. Never present rumours as confirmed facts.

Legal and ethical concerns If the system surfaces genuinely non-public insider information, there are potential legal implications for users acting on it. Mitigation: The system only surfaces publicly posted social media content. It does not access private communications. All rumour alerts clearly state the source is unverified social media speculation. A disclaimer is displayed on every output stating that nothing constitutes financial or legal advice.

Platform dependency The system is currently implemented on Reddit only. API terms, rate limits, or policy changes could disrupt data access. Mitigation: The platform-agnostic architecture means the Collector Agent can be swapped for a different data source with minimal changes to downstream agents.

Market manipulation Coordinated pump-and-dump campaigns can artificially skew sentiment on Reddit. Mitigation: Weight by upvote_ratio. Flag posts with anomalous engagement patterns for manual review.

Data staleness Batch processing means insights may lag real-time market events by hours. Mitigation: Display the data timestamp clearly in every report. State explicitly that this is not a real-time system.

---

11. Short Team Plan

Component	Lead
Data pipeline - Collector and Filter agents	Farrukh
LLM integration - Sentiment Agent	Farrukh
Aggregation logic - Aggregator Agent	Pablo & Moid
Frontend dashboard - Output Agent	Moid
Evaluation design and test cases	Pablo
Phase documentation and diagrams	Afaq
Final report and 5-minute video	Afaq

Track A Additions

Preliminary agent roles: Five agents with clearly defined roles: Collector, Filter, Sentiment Analyzer, Aggregator, and Output. Each has distinct inputs, outputs, and decision logic as documented in the Agent Canvases above.

Likely tools and APIs: Reddit API via PRAW, Arctic Shift bulk download, Anthropic Claude API (claude-sonnet-4-20250514), pandas, Python, and Streamlit planned for Phase 2.

Why multiple interacting components: The pipeline requires components with fundamentally different capabilities. These include API access, rule-based filtering, LLM reasoning, rumour detection, statistical aggregation, and UI formatting. No single component can do all of these. A single-agent approach would either skip filtering and waste API budget on deleted posts, or skip the LLM step and miss sarcasm entirely. The failure cases from our prototype confirm that each stage must be independently designed and testable. The addition of a rumour detection track with separate routing logic further reinforces the need for distinct coordinating agents.

Multi-Agent System Canvas

Agentic Systems Studio · Phase 1 · Track A

SentimentEdge — A multi-source stock sentiment intelligence platform that transforms raw Reddit posts into ticker-level sentiment reports and insider chatter alerts for retail investors

1 SYSTEM INTENT

Purpose and Use Case (1)

Give retail investors access to stock market signals that have traditionally only been available to institutional players. The system reads thousands of Reddit finance posts and returns a confidence-scored sentiment report and insider chatter alerts for any stock ticker in under 30 seconds.

Target User and Stakeholder (2)

PRIMARY	Retail investors and finance students following Reddit communities
PAIN POINT	Cannot manually read thousands of posts; sarcasm and memes are misleading; no systematic way to detect deal rumours
GOAL	Ticker sentiment + rumour alerts in under 30 seconds, sourced from public social media

2 AGENT PIPELINE

Coordination Pattern and Data Flow (3)

Collector — Reddit / PRAW, Arctic Shift
→ **Filter** — Rule-based noise removal
→ **Sentiment** — Claude API, LLM analysis
→ **Aggregator** — Weighted scoring + routing
→ **Output** — Two-track report

PATTERN	Sequential pipeline. Each agent waits for the previous to complete before executing.
GATE	Filter Agent acts as a hard gate. Low-quality and deleted posts never reach the LLM layer.
ROUTING	Aggregator splits output into two tracks: sentiment report and rumour alert list.
STOP	Pipeline halts when Output Agent delivers the report, or when any agent returns zero results after filtering.

System Inputs (4)

PLATFORM	Reddit (in scope). X, StockTwits, others as future extensions.
SUBREDDITS	r/wallstreetbets, r/investing, r/stocks, r/options, r/stockmarket
DATE RANGE	Configurable. Prototype: Apr 2025 to Mar 2026 (349 days)
USER QUERY	Ticker symbol entered by user at Output stage (e.g. $NVDA)

System Outputs (5)

TRACK 1	SENTIMENT REPORT — Overall score, confidence, emotion breakdown, monthly timeline, post type split, top insights
TRACK 2	RUMOUR ALERT — Unconfirmed deal chatter: claim, source post, rumour type, confidence score, disclaimer
WARNINGS	Low-confidence flag (<0.6), sarcasm-adjustment note (>10% sarcastic posts)

3 TOOLS, DATA AND STATE

Tools and APIs (6)

Data and State Design (7)

Platform Strategy (8)

DATA	Reddit API via PRAW (live), Arctic Shift (historical)
LLM	Anthropic Claude API claude-sonnet-4-20250514
PROCESSING	pandas, regex, Python
FRONTEND	Streamlit dashboard (Phase 2)

RAW DATA	104,398 posts downloaded. 37,269 have body content (35.7%).
LLM BATCH	Top 500 posts by engagement per run. Cost ~$0.50 per run.
STATE	No cross-session persistence. Per-run dataframes only.
OUTPUT	Results saved to CSV. Ticker summaries cached per run.

IN SCOPE	Reddit only for this project
FUTURE	X, StockTwits, LinkedIn — Collector Agent is platform-agnostic by design
WHY REDDIT	Free API, large active finance communities, no $200/month barrier

4 HUMAN OVERSIGHT AND GOVERNANCE

👁️ Human-in-the-Loop Points 9

QUERY	User enters ticker symbol at Output stage. System responds to user intent.
WARNING	Low-confidence outputs prompt user to verify manually before acting.
RUMOUR	All rumour alerts show source post. User judges credibility themselves.

⚖️ Governance and Safety 10

DISCLAIMER	Every output states: not financial or legal advice. Source is public social media only.
RUMOURS	Never presented as confirmed facts. Confidence threshold of 0.7 required before surfacing.
PRIVACY	Only publicly posted content. No private communications accessed.
TIMESTAMP	All reports show data timestamp. Batch system — not real-time.